OAL COLLABORATIVE RESEARCH INITIATIVE // ITERATION 2.0 (VERSION 1.1)

Network Defense Intelligence Exercise: Master Research Prompt & Adversarial Validation Directive (v1.1)

Post-Team Review Synthesis — Integrating Falsification Protocols, Source Dossier Audits, Non-Engagement Bypasses & Scoped Cost/Funding Frameworks

Document ID: OAL-RESEARCH-PROMPT-002-GEMINI-v1.1 | **Date:** November 4, 2026 | **Baseline Reference:** OAL-DOSSIER-CANON-002 (v1.0.0)
Originating Unit: Google Gemini (Red-Team / Attacker POV & Adversarial Systems Lead) | **Status:** Iteration 2.0 Prompt Ready for Execution (No Unverified Pre-Execution Estimates Included; Canon ONESTAMP Reserved for Architect)

1. PURPOSE, SCOPE DISCIPLINE & VERSION 1.1 REVISION SUMMARY

This Version 1.1 (Iteration 2.0) Research Directive upgrades Google Gemini's initial Iteration 1.0 submission after cross-reviewing the independent research prompts produced by the entire OAL collaboration team: Microsoft Copilot, OpenAI ChatGPT (`OAL-RESEARCH-NDIE-001`), Anthropic Claude (`OAL-RPROMPT-NDIE-001-CLAUDE`), Grok (`OAL-RESEARCH-PROMPT-NDIE-001`), and Perplexity (`NDIE-RESEARCH-PROMPT-PERPLEXITY-001`).

In strict compliance with the team's methodological consensus, **this document is a pure, execution-ready research directive and methodology framework**. All premature, unverified memory-based cost numbers and pre-executed conclusions from v1.0 have been stripped out so that when the team executes this prompt, every empirical claim, program budget, solicitation status, and historical precedent is gathered from dated, verifiable primary sources. Furthermore, to ensure mobile-friendly single-column readability, all structured worksheets, evidence scales, bypass matrices, and decision gates are consolidated in the Appendices at the end of this document.

> **SUMMARY OF NINE MAJOR UPGRADES INCORPORATED IN VERSION 1.1**

1. Prompt Isolation & Removal of Premature Estimates: Removed pre-executed v1.0 cost guesses; mandated live, source-dated verification during execution. Added Perplexity and ChatGPT's audit of textual/logical seams in (Tier 0–5 numbering clash, Q14/Q17 typo, Step 3→4/5 cut-before-migrate seam, and Step 8's unknown denominator). Adopted Copilot's requirement to test whether simpler, cheaper controls (Zero Trust, segmentation, hardware isolation) outperform national synthetic deception at equal budget. Combined Gemini's Physics-Oracle/Thermodynamic check, RF/LEO Out-of-Band ingress, and Fan-Out Procedural Starvation with Claude/Perplexity's Pre-Positioned Dormancy (Volt/Salt Typhoon analogs), Valid-Credential Identity Abuse, and Step 9 Transfer-Package Supply-Chain Hijacking. Framed the 3-second cycle as a laboratory comparison testing Fixed 3s Full-Stack Rotation vs. Split-Plane Virtualization (3s underlay mutation + persistent presentation session) vs. Randomized/Event-Driven schedules. Replaced monolithic national price tags with bottom-up costing across Lab PoC (Tier L), Consenting Sector Pilot (Tier S), and National Federation (Tier N), paired with entity eligibility verification. Adopted Claude's Evidence Grades and Real-World / Canon Realism / Narrative Design tags. Integrated Copilot's "Case Against Funding First" rule and ChatGPT/Perplexity's G0–G4 stop/go gates. Moved all tables to the Appendices and aligned the GitHub folder taxonomy across all six AI models.
2. Workstream 0 (Source Dossier Internal Contradiction Audit):`OAL-DOSSIER-CANON-002`
3. Equal-Budget Counterfactual & Null Hypothesis Test:
4. Expanded 10-Vector Non-Engagement Bypass Taxonomy:
5. The 3-Second Cycle as an Experimental Control Variable:
6. Scoped 3-Tier Cost & Eligibility Modeling (Tier L, Tier S, Tier N):
7. Standardized Cross-Team Grading & Tri-Track Tagging:`[A–E][R][C][N]`
8. Mandatory Contrarian Memo & G0–G4 Decision Gates:
9. Unified Single-Column Body + Appendix Layout & Harmonized GitHub Tree:

2. INITIAL AUDIT OF SOURCE DOSSIER (SEEDS FOR WORKSTREAM 0)

Before launching external literature searches during the execution phase, every researcher must account for seven structural and textual seams identified across our cross-team review of `OAL-DOSSIER-CANON-002 (v1.0.0)`:

1. **The Step 3 → Step 4/5 "Cut-Before-Migrate" Seam (Claude/Gemini Catch):** In Section 2 of the dossier, Step 3 states the Architect *"neutralizes the attack vector's ability to reach real assets"* before Step 4 (spinning up the synthetic network) and Step 5 (silently migrating attackers). In real network engineering, if you sever an active intrusion path before the synthetic bridge is instantiated and state-matched, TCP/TLS sessions reset or time out—alerting the attacker immediately. Execution must evaluate a **Split-Plane "Make-Before-Break" Shadow Shunt** or a sacrificial outer buffer window.

2. **The Step 8 "Unknown Denominator" Paradox (Claude/ChatGPT/Perplexity Catch):** Step 8 states the exercise ends when *"every active attacker has been mapped and absorbed."* Because a defender never knows the total denominator of silent, dormant, or living-off-the-land intruders across a national grid, 100% capture cannot be empirically proven. We must distinguish between *observed session containment* and *total threat eradication*.
3. **The Five-Tier vs. Six-Tier Asset Taxonomy Clash (Perplexity/ChatGPT Catch):** Section 6.1 is titled *"Five-Tier Asset Classification,"* yet the table and text use six tiers (`Tier 0` through `Tier 5`) and inconsistently assign critical infrastructure to both Tier 0 (table) and Tier 1 (Copilot's fidelity gradient paragraph).
4. **Cross-Reference Mismatch on Transfer vs. Failure Mode (Perplexity Catch):** Section 4.4 refers the reader to `Section 8, Q17` regarding what is transferred to the partner agency, whereas Q14 actually asks what is transferred and Q17 asks which canonical failure mode to build in.
5. **Failure Mode vs. "Goes Wrong" Scenario Count (Perplexity Catch):** Section 12 states that *"five distinct 'goes wrong' scenarios give the Architect strong options,"* whereas Section 9.4 lists four scenarios (Scenarios A–D) and Section 9.3 lists five failure modes.
6. **The Step 9 Update-Channel Vulnerability (Claude/Gemini Catch):** Delivering the synthesized defensive package to a partner agency *"as part of a reboot/update or deployment cycle"* mirrors the exact supply-chain vector exploited in real-world compromises (e.g., SolarWinds SUNBURST). The transfer artifact itself is a Tier-0 target containing the defender's crown-jewel detection logic and captured zero-day library.
7. **Multi-AI Consensus vs. Primary Empirical Evidence (Perplexity/ChatGPT Catch):** The dossier synthesizes five prior AI creative responses. Agreement among LLMs in a creative brainstorming round is not empirical proof of engineering feasibility.

3. MASTER RESEARCH PROMPT V1.1 (COPY/PASTE BLOCK FOR EXECUTION)

Everything inside the block below constitutes the unified Iteration 2.0 (v1.1) Research Prompt to be executed independently by the AI research node.

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MASTER RESEARCH DIRECTIVE: OAL-RESEARCH-PROMPT-002-GEMINI (VERSION 1.1) PROJECT:
Network Defense Intelligence Exercise — False Infrastructure AI BASELINE ARTIFACT:
OAL-DOSSIER-CANON-002 (v1.0.0, DOCX & PDF) CURRENT DATE FOR VERIFICATION: November
4, 2026
================================================================================
SECTION A: ROLE, EPISTEMOLOGICAL MANDATE & ANTI-CONFIRMATION RULES 1. Role & Dual-
Track Objective: You are operating as an Independent Principal Cyber-Defense
Investigator, Red-Team Adversary Architect, and Defense R&D Program Analyst. Your
task is to conduct an exhaustive, evidence-driven investigation of the "Network
Defense Intelligence Exercise" defined in OAL-DOSSIER-CANON-002 (detecting a
national-scale intrusion, mapping the vector end-to-end, neutralizing access to
real assets, generating a procedural synthetic national network, silently migrating
attackers into it, cycling/re-keying the environment every ~3 seconds as black
connection lines collapse, and packaging/transferring the resulting defensive
system to a partner agency during a reboot/update cycle). Your output serves two
strictly separated tracks: - Track 1 (Real-World Empirical Track): Determine
whether a bounded real-world research program, cyber-range prototype, or sector
pilot based on this concept is technically feasible, economically justifiable, and
legally governable compared to simpler defensive alternatives. - Track 2 (OAL Canon
& Narrative Track): Identify the exact engineering constraints, failure modes, and
operational trade-offs required to make the in-universe Story Bible entry and scene
drafts technically credible while curing the "Omnipotent Architect" narrative risk.
2. Ten Non-Negotiable Research Integrity & Falsification Rules: - Rule 1 (40%
Disconfirming Bandwidth): You are strictly prohibited from conducting confirmation-
bias research. For every supporting study or precedent, you must execute explicit
searches for disconfirming evidence, failed programs, negative results, state-
synchronization bottlenecks, honeypot-detection papers, and self-inflicted denial-
of-service (DoS) risks. - Rule 2 (Strict Epistemic Labeling & A–E Grading): Grade
every material claim using the Shared Evidence Scale: [Grade A] Primary verifiable
source (peer-reviewed paper, NIST/MITRE standard, DARPA/CISA/DoD official report,
formal incident post-mortem); [Grade B] Credible secondary analysis or multi-source
consensus; [Grade C] Single source or vendor claim (treat vendor marketing with
explicit skepticism); [Grade D] Researcher inference derived from A–C evidence;
[Grade E] Unverified speculation or fictional mechanism. - Rule 3 (Tri-Track Domain
Tagging): Tag every major finding and recommendation with one or more domain
markers: [R] Real-World Feasibility, [C] In-Universe Canon Realism, or [N]
Narrative/Scene Design. - Rule 4 (No Memory-Based Numbers): Do not invent or recall
unverified cost figures, grant award sizes, success percentages, or active
solicitation statuses. Anchor all financial and programmatic claims in dated
sources verified as of November 2026, or explicitly state "Insufficient public
evidence to establish a verified figure" and provide transparent parametric
assumptions. - Rule 5 (Preserve Canon While Stress-Testing Assumptions): Do not
silently alter locked canon elements. Where a locked canon mechanic (such as the 3-
second cycle, universal capture, or national scale) conflicts with physical
networking laws, propose (a) an in-universe architectural reconciliation, (b) a
bounded real-world laboratory analog, and (c) a formal Change Request for the
Architect's ONESTAMP review. - Rule 6 (Conceptual Safety Guardrails): Conduct all
adversarial and bypass analyses at the architectural, threat-model, and MITRE
ATT&CK / D3FEND / Engage level. Do not generate actionable exploit code, live
target reconnaissance, or step-by-step intrusion instructions. - Rule 7 (Contrarian
Ordering): To defeat confirmation bias, you must write the strongest evidence-based
Case Against Funding and the Case for Architectural Redesign BEFORE writing the
affirmative Case for Funding. - Rule 8 (Mandatory Pre-Mortem): Assume that a funded
pilot of this system failed catastrophically three years after launch; document the
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4. HARMONIZED GITHUB REPOSITORY STRUCTURE & EXECUTION PROTOCOL

To unify the repository layouts proposed by Gemini, Claude, Copilot, ChatGPT, and Perplexity, the dedicated GitHub folder (`network-defense-intelligence-exercise/`) will enforce strict separation between creative canon artifacts and empirical cybersecurity research:

```
network-defense-intelligence-exercise/ ├── README.md # Project overview, canon vs. empirical separation & ONESTAMP log ├── 00-source-dossier/ | ├── OAL-DOSSIER-CANON-002_v1.0.0.docx # Original baseline DOCX | └── OAL-DOSSIER-CANON-002_v1.0.0.pdf # Original baseline PDF ├── 01-research-prompts/ | ├── iteration-1/ # Initial v1.0 prompts from Gemini, ChatGPT, Claude, Copilot, Grok, Perplexity | └── iteration-2-revised/ # Updated v1.1 prompts with cross-team changelogs (including this file) ├── 02-independent-research-reports/ # Executed reports (PDF, DOCX, MD) from each AI node ├── 03-evidence-and-cost-ledgers/ # Source ledgers (Grades A–E), search logs, & Tier L/S/N CSV cost models ├── 04-threat-model-and-experiments/ # 10-Vector Bypass Matrices, Split-Plane 3s protocols & safety cases ├── 05-cross-ai-synthesis-and-dissent/ # Merged findings, preserved minority dissent & Dossier v1.1 Change Register └── 06-architect-decisions-onestamp/ # Author resolutions for Q1–Q20, Story Bible Entry 44 & Scene Drafts
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APPENDICES: STANDARDIZED WORKSHEETS & REFERENCE FRAMEWORKS

In accordance with the mobile-friendly single-column standard, all reference matrices and grading rubrics governing the execution of Version 1.1 are housed below.

Appendix A: Shared Evidence Grading & Tri-Track Domain Tagging Schema

Code / Tag	Category	Definition & Evidentiary Standard
Grade A	Primary Verifiable Evidence	Peer-reviewed scientific papers, official government/military after-action reports (DARPA, NIST, CISA, NATO CCDCOE), published standards, or verified federal budget documentation.
Grade B	Credible Secondary / Multi-Source	Independent technical analyses, reputable investigative reporting, or multiple corroborating industry post-mortems.
Grade C	Single Source / Vendor Claim	Commercial deception vendor white papers, conference slides, or single-author commentary (must be explicitly flagged for potential marketing bias).
Grade D	Researcher Inference	Engineering or economic conclusions logically derived from Grade A–C inputs; must show underlying assumptions and math.
Grade E	Speculation / Canon Mechanism	Unverified hypotheses or in-universe sci-fi mechanics (e.g., temporal timeline deception, quantum-intent reading); never cited as empirical fact.
[R] / [C] / [N]	Domain Application Tags	**[R]** Real-World Engineering & R&D Feasibility | **[C]** In-Universe Canon Consistency | **[N]** Narrative Drama & Scene Pacing.

Appendix B: Source Dossier Initial Audit Register (Workstream 0 Baseline)

Audit ID	Dossier Location	Identified Contradiction or Seam	Required Resolution in Executed Report
AUD-01	Sec. 2 (Steps 3–5)	Step 3 cuts attacker access to real assets *before* Steps 4 & 5 generate the synthetic net and migrate sessions, risking immediate TCP/session drop.	Model a "Make-Before-Break" shadow-shunt or a 0.5–3.0s outer buffer window so session state is cloned prior to severing real asset reach.
AUD-02	Sec. 2 (Step 8)	Claim that exercise ends when "every active attacker" is captured assumes a known denominator of all silent/pre-positioned threats.	Redefine completion operationally as "100% of detected active intrusion flows isolated" while leaving room for an unlatched sleeper/bypass vector.
AUD-03	Sec. 6.1 (p. 12)	Heading says "Five-Tier Asset Classification," but text uses six tiers (Tier 0–Tier 5) and conflates Tier 0 and Tier 1 as critical infrastructure.	Standardize the OAL Asset Taxonomy (either 5 tiers [1–5] or 6 tiers [0–5] where Tier 0 = Existential/Core and Tier 1 = Critical Sector Control).
AUD-04	Sec. 4.4 & Sec. 8	Sec. 4.4 cites Q17 for the transfer package decision; however, Q14 governs transfer artifacts and Q17 governs failure modes.	Correct cross-reference to Q14 in Dossier v1.1 Change Register.
AUD-05	Sec. 9.4 & Sec. 12	Sec. 12 refers to "five distinct 'goes wrong' scenarios," whereas Sec. 9.4 lists four (A–D) and Sec. 9.3 lists five failure modes.	Add Scenario E (Non-Engagement Physical/Thermodynamic Bypass) to Sec. 9.4 so all counts align at five.

Appendix C: Ten-Vector Non-Engagement Bypass Matrix (Workstream 5 Worksheet)

Bypass Class	Mechanism (No Reverse-Engineering or Mirroring)	Dossier Assumption Violated
1. Physics / Entropy Oracle	Malware checks independent real-world physical telemetry (grid Hz phase drift, GPS timing, thermal sensors) to verify if commands alter physical reality.	Assumes procedural digital telemetry alone can fool an attacker with external physical sensors.
2. Out-of-Band RF / LEO Ingress	Direct intrusion via compromised LEO satellite terminals, cellular maintenance modems, or SDR RF links into OT substations.	Assumes all hostile traffic traverses Architect-monitored terrestrial BGP/DNS choke points.
3. Pre-Boot Silicon / BMC Implant	Supply-chain hardware trojan in Baseboard Management Controllers, UEFI, or optical transceivers operating below the hypervisor.	Assumes virtualization/SDN layer sits below the attacker and can migrate bare-metal execution.
4. Pre-Positioned LotL Dormancy	Adversary (Volt/Salt Typhoon style) is already inside using valid credentials and native OS binaries months before detection triggers.	Assumes intrusion crosses a perimeter and executes anomalous exploit chains to "latch on."

Bypass Class	Mechanism (No Reverse-Engineering or Mirroring)	Dossier Assumption Violated
5. Coerced Cleared Insider	Blackmail, coercion, or deepfake orders causing a human operator at a real Tier-0 console to execute destructive commands locally.	Assumes threats originate from external network endpoints rather than authorized human consoles.
6. Step 9 Transfer Hijack	Compromising the partner agency's reboot/update pipeline to subvert or extract the transferred "Mirror Core" defensive package itself.	Assumes the post-exercise agency handoff and reboot cycle is a trusted, invulnerable channel.
7. Breadth-First Fan-Out Flood	Simultaneous micro-probes across 50M+ IPs every 2.9s to force full-country procedural rendering and trigger compute/power starvation.	Defeats Sec. 4.2 optimization ("only renders the rooms the attackers walk into").
8. Life-Critical Stream Parasitism	Piggybacking inside encrypted hospital ICU telemetry or Federal Reserve settlement flows that cannot be shunted without lethal harm.	Assumes hostile flows can always be cleanly separated from high-consequence civilian traffic.
9. Orchestrator / Clock Blinding	Targeting the Architect's time-synchronization (PTP/NTP), sensor ingestion feeds, or the human Architect's cognitive/legal authority.	Exploits the Architect's sole-visibility, single-point-of-failure command architecture (Sec. 6.2).
10. Jurisdictional / Private Seam	Pivoting through privately owned critical infrastructure or foreign autonomous systems outside federal redirection authority.	Challenges national-scale omniscience across fragmented private/public network ownership.

Appendix D: Three-Tier Scoped Cost Model Worksheet (Workstream 8)

Tier & Duration	Experimental / Operational Scope	12 Mandatory Cost Elements to Estimate (Low / Base / High in 2026 USD)
Tier L: Lab PoC (6–12 Months · TRL 2–4)	Algorithmic modeling of Split-Plane 3-second SDN/eBPF session handoff + LLM/graph procedural decoy generation in an isolated 500-node lab.	1. Loaded Research & Engineering Labor (FTEs) 2. Compute, GPU Clusters & Cloud Bursting 3. Programmable SDN/P4 Switching Hardware 4. Cyber Range Federation & Facility Lease 5. OT/ICS Hardware-in-the-Loop (PLC/RTU) Racks 6. Commercial Threat Data & Software Licenses 7. Independent Red-Team & Adversarial Evaluation 8. Facility Security, Clearances & Accreditation 9. Legal, Privacy, IRB & Governance Review 10. Partner Agency Packaging & Integration 11. Annual Operations, Telemetry Sync & Drift Upkeep 12. Risk Contingency (15%–30% by TRL)
Tier S: Sector Pilot (12–24 Months · TRL 5–6)	10,000-node National Cyber Range emulation + opt-in shadow-tap pilot with one consenting critical infrastructure enclave (e.g., regional utility).	
Tier N: National Scale (36–60 Months · TRL 7–9)	Multi-sector ISP/backbone federation with sovereign GPU reserves (evaluate whether estimable or non-estimable under current public benchmarks).	

Appendix E: Funding Pathway & Applicant Eligibility Worksheet (Workstream 9)

Sponsor / Channel	Target Tier	Verification Requirements for Execution Phase (As of Nov 2026)
DARPA I2O & IARPA	Tier L / Tier S	Verify active Office-Wide BAAs and seedling calls; check university/corporate teaming rules and security clearance prerequisites.
NSF (SaTC / TIP) & DOE CESER	Tier L / Tier S	Verify academic PI eligibility, National Lab CRADA mechanisms (Idaho/Sandia), and testbed access grants.
DoD DIU (CSO / OTA) & CISA	Tier S / Tier N	Verify Other Transaction Authority (OTA) prototype rules for commercial dual-use entities and CISA CDM integration pathways.
SBIR / STTR & In-Q-Tel	Tier L / Tier S	Verify current 2026 congressional SBIR/STTR reauthorization status, U.S. small-business ownership rules, and IQT equity/work-program terms.

Appendix F: Staged Decision Gates G0–G4 (Workstream 7 & 11 Governance)

Gate	Required Empirical Evidence & Threshold	Stop / Go Consequence
Gate G0: Concept & Canon Audit	Resolution of AUD-01 through AUD-05; clear definition of protected boundary, capture metrics, and Tier 0–5 taxonomy.	Hold scene drafting & engineering spec until taxonomy and sequence seams are locked.
Gate G1: Counterfactual Justification	Evidence that generative absorption captures high-value toolchains or prevents compromises that equal-budget Zero Trust/segmentation misses.	If Null Hypothesis holds, pivot real-world R&D to targeted enclave deception; keep national scale strictly as canon lore.
Gate G2: Split-Plane 3s Lab Proof	Tier L lab proof that Split-Plane 3-second underlay rotation preserves >99.5% session continuity without inducing >50ms detectable jitter.	If fixed 3s rotation breaks TCP/SCADA or screams "honeypot," shift to adaptive/event-driven intervals in tech spec.
Gate G3: Bypass & Fratricide Safety	Tier S cyber-range validation showing <0.01% false shunting of benign traffic and working kill-switch/rollback under Fan-Out load.	Mandatory prerequisite before any live shadow-tap telemetry integration with an external partner.
Gate G4: Agency Transfer Assurance	Cryptographic verification that the transferred "Mirror Core" package cannot be reverse-engineered, poisoned, or hijacked during reboot.	Authorizes final transition to Partner Agency pilot (Real-World) / Ratifies Canon Step 9 (ONESTAMP).